

SHORT-TERM REGULATION OF ARTERIAL BLOOD PRESSURE

Human Physiology - I | Unit III | Cardiovascular System

Source: K. Sembulingam & Prema Sembulingam, Essentials of Medical Physiology, 6th Edition, Chapter 103

Quick definition

Short-term regulation of arterial blood pressure is the rapid nervous control of arterial pressure. It acts within seconds to minutes mainly through the vasomotor system, baroreceptors and chemoreceptors. It is quick but temporary because the nervous mechanism adapts if the altered pressure persists.

1. Why short-term regulation is needed

Arterial blood pressure changes constantly during posture change, exercise, emotion, sleep, blood loss and respiration. Even small sudden falls in pressure can reduce blood flow to vital organs such as the brain and heart. Therefore, the body uses a rapid feedback mechanism to restore the pressure toward normal before long-term renal and hormonal mechanisms become fully effective.

- It is mainly a nervous mechanism and is the fastest regulatory mechanism for arterial blood pressure.
- It operates through the vasomotor system, which modifies vessel diameter, heart rate, force of contraction, cardiac output and peripheral resistance.
- Its most important reflex is the baroreceptor reflex, also called the pressure buffer mechanism.
- The mechanism acts rapidly but adapts over time; therefore it cannot maintain blood pressure for many days by itself.

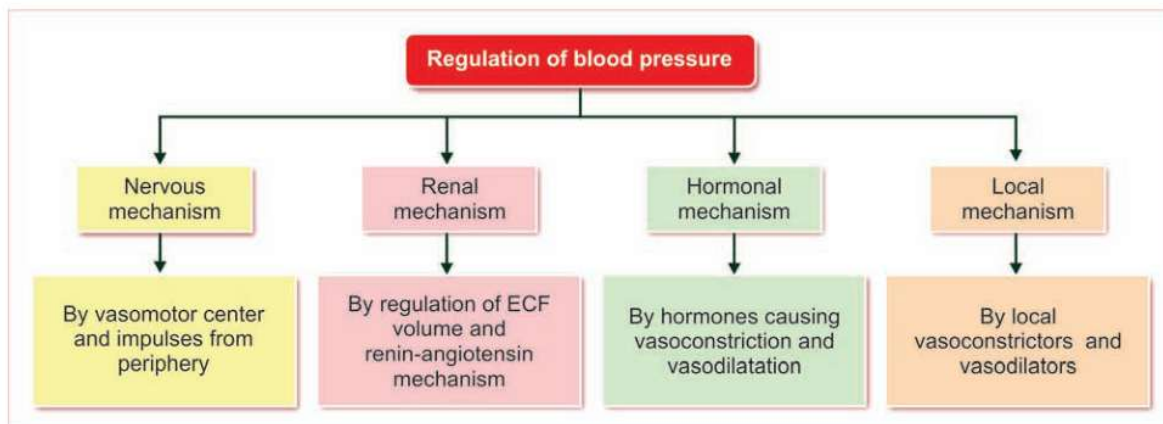


FIGURE 103.2: Regulation of blood pressure. ECF = Extracellular fluid.

Fig. 1: Regulation of blood pressure. The nervous mechanism is the short-term mechanism, while renal regulation is long-term.

2. Mechanisms involved

Mechanism	Stimulus	Main effect	Importance
Baroreceptor mechanism	Stretch of carotid sinus and aortic arch caused by change in arterial pressure	Adjusts sympathetic and parasympathetic output	Most important immediate mechanism; buffers sudden pressure changes
Chemoreceptor mechanism	Low O ₂ , high CO ₂ , high H ⁺ concentration, especially during marked fall in pressure	Stimulates vasoconstrictor centre and raises pressure	Important when pressure falls severely and blood flow to chemoreceptors decreases
Higher centre influence	Emotion, fear, pain, stress, hypothalamic and cerebral cortical impulses	May increase or decrease vasomotor tone	Explains rise of BP during anxiety, stress and emotional states
Respiratory centre influence	Respiratory cycle and thoracic pressure changes	Produces small cyclic BP variations	Causes slight BP rise during early expiration and fall during late expiration/inspiration

3. Vasomotor system

The vasomotor system is the effector system of short-term blood pressure regulation. It controls the tone of arterioles and veins and also influences cardiac activity. It includes the vasomotor centre, vasoconstrictor fibres and vasodilator fibres.

- 1. Vasomotor centre:** the central integrating centre situated bilaterally in the reticular formation of the medulla oblongata and lower pons.
- 2. Vasoconstrictor fibres:** sympathetic fibres that constrict blood vessels mainly by releasing noradrenaline.
- 3. Vasodilator fibres:** fibres that dilate blood vessels in selected regions through parasympathetic, sympathetic cholinergic or antidromic mechanisms.

4. Vasomotor centre

The vasomotor centre is the chief cardiovascular centre concerned with rapid regulation of blood pressure. It receives sensory information from baroreceptors, chemoreceptors, higher centres and respiratory centres, and sends appropriate output to the heart and blood vessels.

Area of vasomotor centre	Other name	Location / role	Effect on BP
Vasoconstrictor area	Pressor area	Forms the lateral portion of vasomotor centre. Sends impulses to blood vessels through sympathetic vasoconstrictor fibres.	Causes vasoconstriction and raises arterial blood pressure. It is also associated with acceleration of heart rate.
Vasodilator area	Depressor area	Forms the medial portion of vasomotor centre. It suppresses the vasoconstrictor area.	Causes vasodilatation and lowers arterial blood pressure. It is also related to cardioinhibition.
Sensory area	Nucleus tractus solitarius area	Receives impulses from periphery through glossopharyngeal and vagus nerves, especially from baroreceptors.	Controls pressor and depressor areas according to incoming sensory information.

Exam tip

In a 5-mark answer, always write the three parts of the vasomotor centre: vasoconstrictor/pressor area, vasodilator/depressor area and sensory area. Then explain baroreceptor reflex with increased BP and decreased BP separately.

5. Vasoconstrictor fibres and vasomotor tone

Vasoconstrictor fibres belong to the sympathetic division of the autonomic nervous system. These fibres release noradrenaline, which acts mainly on alpha receptors of vascular smooth muscle and causes vasoconstriction. The vasoconstrictor fibres play a major role in maintaining normal arterial pressure because arteriolar constriction directly increases peripheral resistance.

Vasomotor tone is the continuous discharge of impulses from the vasoconstrictor centre through sympathetic vasoconstrictor fibres. It keeps the blood vessels in a state of partial constriction. This tone is important because arterial pressure is directly proportional to peripheral resistance. If vasomotor tone increases, vessels constrict and blood pressure rises; if it decreases, vessels dilate and blood pressure falls.

6. Vasodilator fibres

Vasodilator fibres are less dominant than vasoconstrictor fibres in routine blood pressure control, but they are important in special regions such as skeletal muscle and in certain reflex responses.

Type	Neurotransmitter / pathway	Important point
Parasympathetic vasodilator fibres	Release acetylcholine	Cause dilatation of blood vessels in some regions.

Type	Neurotransmitter / pathway	Important point
Sympathetic vasodilator fibres	Sympathetic cholinergic fibres release acetylcholine instead of noradrenaline	Supply skeletal muscle vessels and are important in increasing muscle blood flow, especially during exercise or preparation for exercise.
Antidromic vasodilator fibres	Impulses pass in reverse direction through branches of sensory nerve fibres	Seen in axon reflex; cutaneous receptors may cause local vasodilatation.

7. Baroreceptor mechanism

Baroreceptors are stretch receptors that respond to changes in arterial pressure. They are also called pressoreceptors. The main baroreceptors are located in the carotid sinus and in the wall of the aortic arch. The carotid sinus receptors send impulses through the glossopharyngeal nerve, while the aortic arch receptors send impulses through the vagus nerve. These impulses reach the sensory area of the vasomotor centre, especially the nucleus tractus solitarius.

Nerve from carotid sinus: Glossopharyngeal nerve, through Hering's nerve branch.

Nerve from aortic arch: Vagus nerve.

Central receiving area: Nucleus tractus solitarius in medulla.

Other name of baroreceptor reflex: Pressure buffer mechanism or sinoaortic mechanism.

Diagram checklist for exam

Draw two branches from nucleus tractus solitarius: one branch inhibits the vasoconstrictor area, producing decreased vasomotor tone and dilatation of blood vessels; the second branch stimulates the vasodilator area, producing increased vagal tone, bradycardia and reduced cardiac output. End both branches at normal blood pressure.

When drawing the baroreceptor reflex	Labels you should include
Left side input pathway	Increase in BP -> baroreceptors -> impulses via IX and X cranial nerves -> nucleus tractus solitarius
Vascular pathway	Inhibition of vasoconstrictor area -> decrease in vasomotor tone -> dilatation of blood vessels
Cardiac pathway	Stimulation of vasodilator area -> increase in vagal tone -> bradycardia and reduction in cardiac output
Final correction	Blood pressure returns toward normal

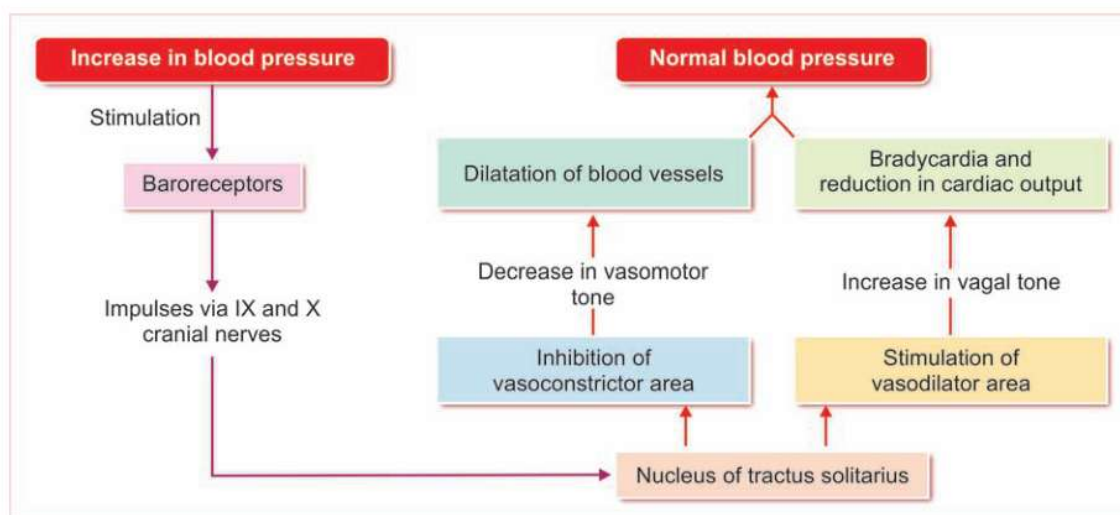


FIGURE 103.3: Regulation of blood pressure by baroreceptor mechanism

Fig. 2: Regulation of blood pressure by baroreceptor mechanism when arterial pressure increases.

8. When arterial blood pressure increases

When arterial pressure rises rapidly, the carotid sinus and aortic arch are stretched more. This increased stretch stimulates the baroreceptors. The frequency of impulses from the baroreceptors increases and the impulses reach the nucleus tractus solitarius through the glossopharyngeal and vagus nerves.

Step	Event
1	Increased BP stretches carotid sinus and aortic arch.
2	Baroreceptors are stimulated and impulse discharge increases.
3	Impulses reach nucleus tractus solitarius via IX and X cranial nerves.
4	Sensory area inhibits vasoconstrictor area and stimulates vasodilator area.
5	Sympathetic vasoconstrictor tone decreases, causing vasodilatation.
6	Vagal tone to heart increases, producing bradycardia and reduced force of contraction.
7	Peripheral resistance and cardiac output decrease.
8	Arterial blood pressure falls back toward normal.

Flow to remember - Increased BP

Increased BP → stretch of baroreceptors → increased impulses via IX and X nerves → nucleus tractus solitarius → inhibition of vasoconstrictor area + stimulation of vasodilator area → vasodilatation + bradycardia → decreased cardiac output and peripheral resistance → BP decreases.

9. When arterial blood pressure decreases

When arterial pressure falls, the stretch on baroreceptors decreases. Therefore the discharge of impulses from the baroreceptors decreases. This removes the inhibitory effect on the vasoconstrictor area, so the pressor area becomes more active. At the same time, vagal tone decreases. The combined effect raises blood pressure.

Step	Event
1	Fall in BP reduces stretch of carotid sinus and aortic arch.
2	Baroreceptor firing decreases.
3	Less impulse reaches the sensory area of vasomotor centre.
4	Vasoconstrictor area is no longer inhibited.
5	Sympathetic activity increases and parasympathetic activity decreases.
6	Arterioles constrict, increasing total peripheral resistance.
7	Heart rate and force of contraction increase, raising cardiac output.
8	Arterial blood pressure rises toward normal.

Flow to remember - Decreased BP

Decreased BP → less stretch of baroreceptors → reduced impulses via IX and X nerves → reduced inhibition of vasoconstrictor area → increased sympathetic tone → vasoconstriction + tachycardia + increased force of contraction → increased peripheral resistance and cardiac output → BP increases.

10. Important features of baroreceptor reflex

- It responds more strongly to rapid changes in pressure than to steady pressure.
- It is very important in postural changes. On standing, BP tends to fall due to venous pooling; the baroreceptor reflex restores pressure by increasing sympathetic activity.
- It is called a pressure buffer because it buffers or minimizes sudden rise or fall in arterial pressure.
- The nerves from baroreceptors are called buffer nerves because they buffer sudden pressure changes.
- It works mainly in the pressure range of normal and moderately altered arterial pressures. With prolonged pressure change, the receptors may reset, so long-term control depends mainly on renal mechanisms.

11. Chemoreceptor mechanism

Chemoreceptors are receptors that respond to changes in the chemical composition of blood. Peripheral chemoreceptors are situated in the carotid body and aortic body. They are especially sensitive to lack of oxygen, excess carbon dioxide and increased hydrogen ion concentration. When blood pressure falls markedly, blood flow to chemoreceptors decreases. This causes accumulation of CO₂ and H⁺ and reduction of O₂ in the chemoreceptor area, which stimulates them.

Stimulus	Pathway	Effect
Low O ₂ , high CO ₂ or high H ⁺ in blood	Carotid body and aortic body send impulses to vasomotor centre	Stimulates vasoconstrictor centre and raises BP
Severe fall in BP	Reduced blood flow to chemoreceptors causes chemical changes	Strong sympathetic vasoconstriction and increase in pressure
Associated respiratory effect	Chemoreceptor stimulation also activates respiratory centres	Improves ventilation and helps correct blood gas changes

This mechanism is less important than the baroreceptor reflex at normal pressures, but it becomes very important during severe hypotension, shock and marked reduction in blood flow. In such conditions, strong vasoconstriction helps maintain blood flow to vital organs.

12. Sinoaortic mechanism

The term sinoaortic mechanism refers to the combined action of baroreceptors and chemoreceptors present in the carotid sinus, carotid body, aortic arch and aortic body. The nerves supplying these receptors are known as buffer nerves because they regulate heart rate, vascular tone, blood pressure and respiration.

- Carotid sinus and aortic arch mainly act as baroreceptor areas.
- Carotid body and aortic body mainly act as chemoreceptor areas.
- Together they help in rapid compensation for sudden changes in pressure and blood gases.

13. Influence of higher centres

The vasomotor centre is not isolated. It receives impulses from higher centres of the brain. This explains why emotional states, fear, pain, anxiety and stress can change blood pressure. The cerebral cortex, especially areas related to emotional reaction, can send impulses to the vasomotor centre and increase vasomotor tone. This increases blood pressure.

Higher centre	Effect
Cerebral cortex	Emotional states can activate vasomotor centre, increase vasomotor tone and raise BP.
Hypothalamus - posterior and lateral nuclei	Stimulation causes vasoconstriction and rise in blood pressure.
Hypothalamus - preoptic area	Stimulation causes vasodilatation and fall in blood pressure.

14. Influence of respiratory centres

Respiration produces small rhythmic changes in arterial pressure. At the beginning of expiration, arterial pressure increases slightly by about 4 to 6 mm Hg. It decreases during late expiration and during inspiration. These changes occur because impulses from respiratory centres influence the vasomotor centre at different phases of the respiratory cycle, and because pressure changes in the thoracic cavity alter venous return and cardiac output.

15. Applied physiology

Situation	Role of short-term regulation
Standing from lying position	Venous pooling in legs tends to reduce BP. Baroreceptor firing falls,

Situation	Role of short-term regulation
	sympathetic tone increases, arterioles constrict and heart rate increases.
Sudden rise in BP	Baroreceptor firing increases. Sympathetic tone decreases and vagal tone increases, causing vasodilatation and bradycardia.
Blood loss / early shock	Reduced BP reduces baroreceptor firing. Sympathetic discharge increases to maintain pressure. Chemoreceptors assist when hypotension is severe.
Carotid sinus massage	Stimulates carotid sinus baroreceptors and increases vagal tone; may reduce heart rate.
Emotional stress or fear	Higher centres activate vasomotor centre and increase sympathetic tone, raising BP.

16. Short answer format for exam

Write in this order

Definition -> nervous mechanism -> vasomotor system -> vasomotor centre areas -> baroreceptor reflex in increased BP and decreased BP -> chemoreceptor mechanism -> higher centres/respiratory centres -> applied importance.

1. Short-term regulation of arterial blood pressure is rapid nervous regulation acting within seconds to minutes.
2. It acts through vasomotor system: vasomotor centre, vasoconstrictor fibres and vasodilator fibres.
3. The vasomotor centre is situated in medulla and lower pons and has pressor, depressor and sensory areas.
4. Baroreceptors are present in carotid sinus and aortic arch; their impulses travel through IX and X cranial nerves to nucleus tractus solitarius.
5. Increased BP stimulates baroreceptors, causing vasodilatation, bradycardia and decreased cardiac output, so BP falls.
6. Decreased BP reduces baroreceptor firing, causing sympathetic activation, vasoconstriction, tachycardia and increased cardiac output, so BP rises.
7. Chemoreceptors in carotid and aortic bodies help during severe fall in BP by stimulating vasoconstrictor centre.
8. Higher centres and respiratory centres also modify vasomotor activity.

17. One-page revision table

Point	Must remember
Fastest mechanism	Nervous mechanism
Main centre	Vasomotor centre in medulla oblongata and lower pons
Main reflex	Baroreceptor reflex
Main receptors	Carotid sinus and aortic arch baroreceptors
Afferent nerves	Glossopharyngeal nerve from carotid sinus; vagus nerve from aortic arch
Receiving nucleus	Nucleus tractus solitarius
In increased BP	Increased baroreceptor firing -> vasodilatation + bradycardia -> BP decreases
In decreased BP	Decreased baroreceptor firing -> sympathetic activation -> vasoconstriction + tachycardia -> BP increases
Chemoreceptors	Carotid body and aortic body; respond to low O ₂ , high CO ₂ and high H ⁺
Limitation	Adapts/reset with prolonged pressure change; not enough for long-term BP control

Common mistakes to avoid

Do not write only baroreceptors. For a complete answer, include vasomotor centre, vasoconstrictor fibres, vasodilator fibres, baroreceptors, chemoreceptors and higher-centre influence. Also remember: increased BP increases baroreceptor firing, but decreased BP decreases baroreceptor firing.

